

Experiment 4

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Subject Name: Full Stack Development

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Aim:

To optimize the performance of the EcoTrack React application using memoization techniques and code splitting, and to enhance the user interface using enterprise-grade Material UI components.

Objective:

After completing this experiment, the student will be able to:

- Understand the causes of unnecessary re-renders in React applications
- Optimize React components using React.memo to prevent avoidable re-renders
- Apply useMemo to efficiently compute derived data and avoid redundant calculations
- Use useCallback to memoize event handler functions and improve component performance
- Implement lazy loading of components and routes using React.lazy and Suspense
- Reduce initial bundle size and improve application load performance through code splitting
- Enhance the visual appearance and usability of the EcoTrack application using Material UI components
- Design a clean, consistent, and responsive user interface using Material UI layouts and typography

Implementation/Code:

Login.jsx:

```
import { useAuth } from "../context/AuthContext";  
import { useNavigate } from "react-router-dom";  
import Button from '@mui/material/Button';  
import Box from '@mui/material/Box';  
import Typography from '@mui/material/Typography';
```

```
const Login = () => {  
  const { login } = useAuth();  
  const navigate = useNavigate();
```

```
const handleLogin = () => {
  login();
  navigate("/dashboard");
};

return (
  <Box sx={{
    display: 'flex',
    flexDirection: 'column',
    alignItems: 'center',
    justifyContent: 'center',
    minHeight: '100vh',
    p: 3
  }}>
    <Typography variant="h4" gutterBottom>
      Welcome to EcoTrack
    </Typography>
    <Button
      variant="contained"
      color="primary"
      size="large"
      onClick={handleLogin}
      sx={{ mt: 3, px: 6, py: 1.5 }}
    >
      Login to EcoTrack
    </Button>
  </Box>
);
};
```

export default Login;

Dashboard.jsx:

```
import { useState } from 'react';
import { Outlet, useNavigate, Link as RouterLink } from 'react-router-dom';
import { useAuth } from "../context/AuthContext";
import Header from "../components/Header";
```

```
import AppBar from '@mui/material/AppBar';
import Box from '@mui/material/Box';
import Toolbar from '@mui/material/Toolbar';
import Typography from '@mui/material/Typography';
```

```
import Drawer from '@mui/material/Drawer';
import List from '@mui/material/List';
import ListItem from '@mui/material/ListItem';
import ListItemButton from '@mui/material/ListItemButton';
import ListItemIcon from '@mui/material/ListItemIcon';
import ListItemText from '@mui/material/ListItemText';
import Divider from '@mui/material/Divider';
import IconButton from '@mui/material/IconButton';
import Button from '@mui/material/Button';

import HomeIcon from '@mui/icons-material/Home';
import DashboardIcon from '@mui/icons-material/Dashboard';
import AssessmentIcon from '@mui/icons-material/Assessment';
import LogoutIcon from '@mui/icons-material/Logout';
import MenuIcon from '@mui/icons-material/Menu';
```

```
const drawerWidth = 240;
```

```
const Dashboard = () => {
  const { logout } = useAuth();
  const navigate = useNavigate();
  const [mobileOpen, setMobileOpen] = useState(false);
```

```
  const handleLogout = () => {
    logout();
    navigate("/login");
  };
};
```

```
const handleDrawerToggle = () => {
  setMobileOpen(!mobileOpen);
};
```

```
const drawerContent = (
  <div>
    <Toolbar /> {/* spacer under AppBar */}
    <Divider />
    <List>
      <ListItem disablePadding>
        <ListItemButton
          component={RouterLink}
          to=""
          onClick={() => setMobileOpen(false)}
        >
```

```

        <ListItemIcon><HomeIcon /></ListItemIcon>
        <ListItemText primary="Home" />
      </ListItemButton>
    </ListItem>

    <ListItem disablePadding>
      <ListItemButton
        component={RouterLink}
        to="overview"
        onClick={() => setMobileOpen(false)}
      >
        <ListItemIcon><DashboardIcon /></ListItemIcon>
        <ListItemText primary="Overview" />
      </ListItemButton>
    </ListItem>

    <ListItem disablePadding>
      <ListItemButton
        component={RouterLink}
        to="reports"
        onClick={() => setMobileOpen(false)}
      >
        <ListItemIcon><AssessmentIcon /></ListItemIcon>
        <ListItemText primary="Reports" />
      </ListItemButton>
    </ListItem>
  </List>
  <Divider />
  <List>
    <ListItem disablePadding>
      <ListItemButton onClick={handleLogout} sx={{ color: 'error.main' }}>
        <ListItemIcon><LogoutIcon color="error" /></ListItemIcon>
        <ListItemText primary="Logout" />
      </ListItemButton>
    </ListItem>
  </List>
</div>

);

return (
  <Box sx={{ display: 'flex' }}>
    { /* AppBar at top */ }
    <AppBar

```

```

position="fixed"
sx={{
  width: { sm: `calc(100% - ${drawerWidth}px)` },
  ml: { sm: `${drawerWidth}px` }, // ← important: push content right
  backgroundColor: '#4CAF50',
}}
>
<Toolbar>
  <IconButton
    color="inherit"
    edge="start"
    onClick={handleDrawerToggle}
    sx={{ mr: 2, display: { sm: 'none' } }}
  >
    <MenuIcon />
  </IconButton>
  <Typography variant="h6" noWrap component="div">
    Dashboard
  </Typography>
</Toolbar>
</AppBar>

{/* Sidebar - LEFT side */}
<Box
  component="nav"
  sx={{
    width: { sm: drawerWidth },
    flexShrink: { sm: 0 }
  }}
>
  {/* Mobile temporary drawer */}
  <Drawer
    variant="temporary"
    open={mobileOpen}
    onClose={handleDrawerToggle}
    ModalProps={{ keepMounted: true }}
    sx={{
      display: { xs: 'block', sm: 'none' },
      '& .MuiDrawer-paper': {
        boxSizing: 'border-box',
        width: drawerWidth,
        backgroundColor: '#4CAF50', // green theme
        color: 'white',

```

```
    },
  }}
>
  {drawerContent}
</Drawer>
```

```
{/* Desktop permanent drawer - LEFT */}
<Drawer
  variant="permanent"
  sx={{
    display: { xs: 'none', sm: 'block' },
    '& .MuiDrawer-paper': {
      boxSizing: 'border-box',
      width: drawerWidth,
      backgroundColor: '#4CAF50',
      color: 'white',
    },
  }}
  open
>
  {drawerContent}
</Drawer>
</Box>
```

```
{/* Main content - pushed to right */}
<Box
  component="main"
  sx={{
    flexGrow: 1,
    p: 3,
    width: { sm: `calc(100% - ${drawerWidth}px)` },
    ml: { sm: `${drawerWidth}px` }, // ← ensures shift
    mt: '64px', // space under AppBar (Toolbar height)
    backgroundColor: 'background.default',
  }}
>
  {/* Spacer for fixed AppBar */}
  <Toolbar />
```

```
{/* This is where your child routes render: DashboardHome / Overview / Reports
*/}
  <Outlet />
</Box>
```

```

    </Box>
  );
};

export default Dashboard;

```

App.jsx:

```

import { lazy, Suspense } from "react";
import { Routes, Route, Navigate } from "react-router-dom";
import ProtectedRoute from "../routes/ProtectedRoute";

// Lazy load pages
const Dashboard = lazy(() => import("../pages/Dashboard"));
const DashboardHome = lazy(() => import("../pages/DashboardHome"));
const Overview = lazy(() => import("../pages/Overview"));
const Reports = lazy(() => import("../pages/Reports"));
const Login = lazy(() => import("../pages/Login"));

const App = () => {
  return (
    <Suspense fallback={
      <div style={{
        display: 'flex',
        justifyContent: 'center',
        alignItems: 'center',
        minHeight: '100vh'
      }}>
        Loading application...
      </div>
    }>
      <Routes>
        <Route path="/login" element={<Login />} />

        <Route
          path="/dashboard"
          element={
            <ProtectedRoute>
              <Dashboard />
            </ProtectedRoute>
          }
        />
        <Route index element={<DashboardHome />} />
        <Route path="overview" element={<Overview />} />
      </Routes>
    </Suspense>
  );
};

```

```
    <Route path="reports" element={<Reports />} />  
  </Route>
```

```
    <Route path="*" element={<Navigate to="/dashboard" />} />  
  </Routes>
```

```
</Suspense>
```

```
);
```

```
};
```

```
export default App;
```

Main.jsx:

```
import React from "react";  
import ReactDOM from "react-dom/client";  
import App from "./App";  
import { BrowserRouter } from "react-router-dom";  
import { AuthProvider } from "./context/AuthContext";  
import { Provider } from "react-redux";  
import { store } from "./redux/store";  
import { ThemeProvider } from '@mui/material/styles';  
import CssBaseline from '@mui/material/CssBaseline';  
import theme from "./theme";  
import "./index.css";  
import "@fontsource/roboto/300.css";  
import "@fontsource/roboto/400.css";  
import "@fontsource/roboto/500.css";  
import "@fontsource/roboto/700.css";
```

```
ReactDOM.createRoot(document.getElementById("root")).render(  
  <React.StrictMode>
```

```
    <Provider store={store}>
```

```
      <ThemeProvider theme={theme}>
```

```
        <CssBaseline />
```

```
        <BrowserRouter>
```

```
          <AuthProvider>
```

```
            <App />
```

```
          </AuthProvider>
```

```
        </BrowserRouter>
```

```
      </ThemeProvider>
```

```
    </Provider>
```

```
  </React.StrictMode>
```

```
);
```


Output:

Welcome to EcoTrack

Login to EcoTrack

Dashboard

 Home

 Overview

 Reports

 Logout

Total Activities

- Car Travel: 4 Kg
- Electricity Usage: 6 Kg
- Cycling: 0 Kg

High Carbon (> 4 Kg)

- Electricity Usage

Low Carbon ($\leq$ 4 Kg)

- Car Travel
- Cycling

Refresh Logs

Overview Page

This is the Overview section of EcoTrack.

Reports & Analytics

Detailed reports and carbon footprint analysis coming soon...

Learning Outcome:

1. Identify and analyze causes of unnecessary re-renders in a React application.
2. Apply React.memo, useMemo, and useCallback to optimize component rendering performance.
3. Implement code splitting and lazy loading using React.lazy and Suspense to improve load time.
4. Design a responsive and consistent UI using Material UI components and theming.
5. Integrate performance optimization techniques with secure routing and modern React architecture.